# Infection and response

## Cause & spread

### 1 Pathogens cause disease

Bacteria may release toxins; viruses reproduce inside cells. Fungi and protists also cause disease.

Spread: contact, water, air, food or vectors. Hygiene, isolation, vaccination and vector control break routes.

### 2 Named human infections

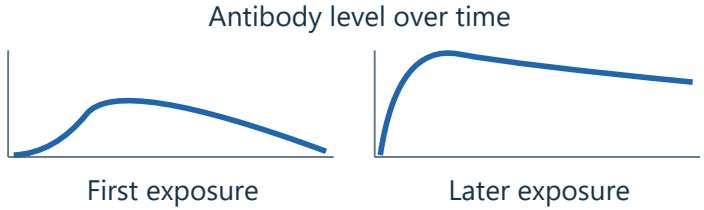
Measles: droplets, fever/rash. HIV: body fluids, damages immune cells; antiretrovirals control it.

Salmonella: food, gut illness. Gonorrhoea: sexual contact, discharge/pain; antibiotic resistance matters.

Malaria: protist carried by mosquitoes; recurrent fever. Nets and breeding control reduce spread.

## Protect & treat

### 3 Defence & vaccination




Skin, mucus/cilia and stomach acid form barriers. White cells engulf pathogens, make antibodies and antitoxins.

Vaccination → specific antibodies/memory → faster later response. High uptake reduces transmission.

### 4 Medicines & targeting

Antibiotics treat bacteria, not viruses. Painkillers treat symptoms. Trials check toxicity, efficacy and dose.

Monoclonals: lymphocyte + tumour cell → hybridoma → clone. Specific binding enables tests and targeted treatments; side effects remain.

## Plant health

### 5 Plant problems

TMV: mosaic leaves; black spot: fungal spots/leaf loss. Both reduce photosynthesis. Aphids feed on sap.

Nitrate shortage → less protein/growth; magnesium shortage → less chlorophyll/chlorosis.

Identify using symptoms, references, labs or antibody tests.

### 6 Plant protection

Cell walls, waxy cuticle and shedding bark resist entry.

Antibacterial chemicals attack bacteria; poisons, thorns and hairs deter feeding.

Drooping/curling leaves and mimicry discourage herbivores.